

Optimal Routing and Design of Radial Medium Voltage Power Distribution Networks Considering Cables Short Circuit Withstand-Capacity

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Abstract—Cables short circuit withstand-capacity plays a crucial role in medium voltage distribution system design. For a given short circuit level at substation, fault current at any point in a radial distribution system depends on the location with respect to the substation, distributed generation as well as size and length of upstream cable sections. This paper addresses optimal routing, cable sizing, distributed generation allocation and sizing of radial medium voltage power distribution network considering cables short circuit withstand capacity. Both synchronous-based and inverter-based distributed generators are considered. Complete formulation of cables short circuit constraints (SCCs) is introduced in mixed-integer linear programming (MILP) form. A Benders-decomposition-based approach is used to solve the problem by considering short circuit at each system bus in a separate slave feasibility problem. Combinatorial cuts are proposed to exclude infeasible solutions in each iteration by investigating upstream routes, cable sizes and distributed generation. The proposed optimization framework and solution algorithm are applied on 54 bus, 84 bus and 138 bus systems. The results revealed potential saving in investment, maintenance and losses costs compared to upgrading method. Also the proposed solution algorithm showed great saving in the number of added constraints to the original problem.

Index Terms—Distribution system planning, mixed integer linear programming (MILP), cables short circuit withstand capacity, benders decomposition, combinatorial cuts.

NOMENCLATURE

A. Sets/Indices

B, B_L	Set of buses and load buses.
B_S, B_G	Set of substation buses and generator candidate buses.
B_{SC} / b	Set /index of short circuit buses.
C / c	Set/index of candidate cable sizes.
D / d	Set/index of load levels.
DR_{bt}	Set of routes just downstream of bus b in iteration t with violated short circuit constraints.

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G / g

i, j, k, α, β

N_b

UB_{bt}, UR_{bt}

Y

t

ν

B. Constants

$C_c^{C, I}, C_g^{G, I}$

$C_d^{S, E}, C_g^{G, E}$

$C_c^{C, M}, C_g^{G, M}$

$\bar{I}_c^C, \bar{I}_0^{SC}$

P_i^L, Q_i^L

l_{ij}

LT

M

n_v

pf_g

q

r_c^C, r_i^S, r_g^G

$\bar{S}^S, \bar{S}_g^G$

U

$V_{\max}, V_{\min}$

x_c^C, x_b^S, x_g^G

z_c^C, z_b^S, z_g^G

μ_d

Set/index of candidate generators.

Indices of buses.

Set of buses connected to bus b .

Set of buses and routes upstream of bus b in iteration t .

Set of available candidate routes.

Index of iteration.

Index of linearization segment.

Investment cost of cable c (USD/Km) and of generator g (USD/kW).

Energy cost of substation at load level d and of generator g (USD/kWh).

Maintenance cost of cable c and of generator g (USD/year).

Ampacity of cable c (kA).

Short circuit capacity of cable c and maximum short circuit level in the system (kA).

Active and reactive load at bus i (MW, MVar).

Length of route ij (km).

Project lifetime (year).

Disjunctive large constant.

Number of linearization segments.

Rated power factor of generator g .

Interest rate.

Resistance of cable c (Ω / km), of substation transformer at bus i and of generator g (Ω).

Capacity of substation and capacity of generator g (kVA).

Pre-fault voltage (kV).

Maximum and minimum voltage limits (kV).

Reactance of cable c (Ω / Km), of substation transformer at bus b and of generator g (Ω).

Impedance of cable c (Ω / Km), of substation transformer at bus b and of generator g (Ω).

Load factor of load level d .

Δ_d	Duration of load level d (Hours).
ρ_{max}	Maximum allowable DG penetration level.

C. Discrete Variables

y_{ijc}^C, y_{ij0}^C	Decision variable for installing cable c and for installing no cables in route ij .
w_{ij}^R	Decision variable for installing cables in route ij with bus i upstream of bus j .
h_{ig}^G, h_{i0}^G	Decision variable for installing generator g and for installing no generators in bus i .

D. Continuous Variables

$I_{bij}^{R, re}, I_{bij}^{R, im}$	Real and imaginary components of short circuit current in route ij for short circuit at bus b (kA).
$I_{bi}^{S, re}, I_{bi}^{S, im}$	Real and imaginary components of short circuit current supplied from substation at bus i for short circuit at bus b (kA).
$I_{bi}^{G, re}, I_{bi}^{G, im}$	Real and imaginary components of short circuit current supplied from generator at bus i for short circuit at bus b (kA).
$I_{bi}^{WC, re}, I_{bi}^{WC, im}$	Real and imaginary components of worst-case short circuit current in cable installed in route bi for short circuit at bus b (kA).
I_{ijcd}^{sq}	Square of the magnitude of current in cable c installed in route ij at load level d (kA) ² .
$P_{ijcd}^C, P_{id}^S, P_{igd}^G$	Active power flow in cable c installed in route ij , supplied from substation at bus i and supplied from generator g installed at bus i for load level d (MW).
$Q_{ijcd}^C, Q_{id}^S, Q_{igd}^G$	Reactive power flow in cable c installed in route ij , supplied from substation at bus i and supplied from generator g installed at bus i for load level d (MVar).
V_{id}^{sq}	Square of voltage magnitude at bus i for load level d .
V_{bi}^{re}, V_{bi}^{im}	Real and imaginary components of voltage at bus i for short circuit at bus b (kV).
$\lambda_{bt}^{EQ}, \lambda_{bt}^{IN}$	Lagrange multipliers of equality and inequality constraints of slave sub problem of short circuit bus b in iteration t .

I. INTRODUCTION

DISTRIBUTION system planning is a techno-economic problem that has always attracted attention of Engineers and researchers [1]. Distribution planning incorporates determination of the best routes of distribution lines, the optimum conductor sizes and optimal allocation and sizing of distributed generation (DG). Mathematically, Distribution system planning is a mixed-integer optimization problem for cost minimization while satisfying system technical and operational constraints [2]–[4].

In medium voltage distribution cable network with predetermined routes, it is well known that short circuit capacity plays the principal rule in cable sizing. The presence of DG also increases short circuit levels depending on location, size

and type. If network routing, cable sizing and DG allocation are all included the planning process, the problem will be more complicated. For example, branching cable sections near the substation may reduce total cabling length but large cable sizes are required to meet the high short circuit level. If the size of an upstream cable is increased in an attempt to meet their high short circuit level, this will increase also the short circuit level for the downstream cables.

In transmission network planning studies like [5]–[7], bus impedance matrix-based short circuit constraints are included as part of transmission system planning but only short circuit level at buses is considered to ensure its compatibility with circuit breaker breaking capacity. Short circuit withstand-capacity of lines or cables has not been considered at all. Also, the included short circuit constraints in [5]–[7] are only approximation that benefits from the characteristics of the transmission network with its high x/r ratio which is not suitable for the distribution network. In [8], constraints for branch circuits short circuit capacity are given but for a fixed network and fixed cable sizes. The optimization variables were only the DG type, size and location. Also worst-case short circuit near upstream buses has not been considered.

Comprehensive overview of the types of constraints involved in distribution expansion planning literature can be found in [1]. To the best of the author's knowledge, short circuit capacity constraints have not been considered in distribution system network planning literature. The common practice in distribution engineering is to perform short circuit study on the obtained optimal network and then upgrade the cables that violate their short circuit capacity by larger sizes starting from the substation and going downstream, but this approach does not guarantee optimal solution.

Heuristic Optimization methods have been extensively applied to distribution system planning [9], [10] but the quality of the obtained solution is not guaranteed. On the other hand, mathematical programming gives an upper bound of the optimality gap which is an indication about solution quality. Benders decomposition has been effectively used to solve complex mixed-integer optimization problems and it has been applied in transmission and distribution planning [7], [11].

In this research, complete formulation of short circuit constraints (SCCs) is derived. The constraints depend on decision variables of network routing and cable size selection. Benders decomposition technique is used to solve the complex design problem; the master problem solves the routing and design problem without considering SCCs then the solution is forwarded to multiple slave feasibility subproblems which consider SCCs at all network busses. Moreover, combinatorial cuts are proposed to speed up convergence by excluding non-feasible solutions.

The main contributions of this work are summarized as follows.

- 1) Constraints of short circuit withstand capacity of cables are introduced and derived for distribution system planning problem in mixed-integer linear programming (MILP) form. The constraints incorporate binary decision variables of the problem because routes, cable sizes and

DG allocation are not known in advance. This guarantees that, in the optimal solution, each cable in each decided route with its decided size will withstand the worst short circuit current in its location.

- 2) Complete modeling of network during short circuit at each bus is given as well as worst-case short circuit current in the form of constraints incorporating binary decision variables and considering both resistance and reactance of the cables, generators and substations transformers.
- 3) Combinatorial cuts are proposed to exclude infeasible solutions in Benders decomposition solution methodology iteratively by investigating combinations of upstream routes, cables and distributed generation.
- 4) The effect of renewable inverter-based distributed generation and increase in distributed generation penetration level on short-circuit-constrained optimal distribution planning is studied.

The paper is summarized as follows: Section II provides problem formulation. Short circuit constraints are introduced in Section III. Solution using Benders decomposition is demonstrated in Section IV and combinatorial cuts are proposed in Section V. Simulation results and discussion are given in Section VI and finally conclusions are summarized in Section VII.

II. PROBLEM FORMULATION

A. Objective Function

The objective function to be minimized is the sum of the annualized cost of investment (IC), maintenance (MC), and energy (EC) as shown in (1). Cables and DG investment cost are summed in (2). Equation (3) represents maintenance cost of cables and DG. Energy costs of substations and DG are included in (4). It should be noted that power losses cost is included in EC. Load duration curve is discretized into multiple load levels for calculation of energy cost.

$$\text{Minimize } F = IC + MC + EC \quad (1)$$

$$IC = \frac{q(1+q)^{LT}}{(1+q)^{LT} - 1} \left(\sum_{ij \in Y} \sum_{c \in C} C_c^C \cdot I_{ij}^C y_{ijc}^C + \sum_{b \in B_G} \sum_{g \in G} C_g^G \cdot \bar{S}_g^G p_{fg} h_{bg}^G \right) \quad (2)$$

$$MC = \sum_{ij \in Y} \sum_{c \in C} C_c^{C, M} y_{ijc}^C + \sum_{b \in B_G} \sum_{g \in G} C_g^{G, M} h_{bg}^G \quad (3)$$

$$EC = \sum_{d \in D} \Delta_d \left(\sum_{b \in B_S} C_d^{S, E} P_{bd}^S + \sum_{b \in B_G} \sum_{g \in G} C_g^{G, E} P_{bgd}^G \right) \quad (4)$$

The optimization problem is subjected to the following power flow and radiality constraints.

B. Power Flow Constraints

Equations (5) and (6) represent active/reactive power balance for each bus. Constraints (7) and (8) impose that injected power from substations and generators is limited to substation buses and busses with candidate DGs only; respectively. Active/reactive power flow is allowed only in installed cables as imposed by (9) and (10); respectively. Equation (11) limits the maximum current flow in installed cables. Maximum power limits for substations and installed DG are given by (12) and (13); respectively. Voltage drop along each candidate cable is governed by (14) [8], [12], [13]. The relation between power, voltage and current for each route is given in (15). Equation (15) is nonlinear, for a MILP model, equation (15) is linearized according to [8], [12], [13]. The terms in equation (15) are defined in (16)-(18). Equation (19) constrains the allowed range for buses voltage in normal operation.

The model assumes balanced three-phase system and constant power loads, linearized equation for unbalanced system can be found in [14], [15]. Constraint (20) fixes substation voltages in normal operation which are assumed to be fully regulated using on-load tap changers (OLTC). Equation (21) limits the power injected by DG to the maximum allowable DG penetration level in all loading levels.

$$\sum_{ki \in Y} \sum_{c \in C} P_{kicd}^C - \sum_{ij \in Y} \sum_{c \in C} \left(P_{ijcd}^C + I_{ijcd}^{\text{sqr}} r_c^C l_{ij} \right) + P_{id}^S + P_{igd}^G = \mu_d P_i^L; \forall i \in B, d \in D \quad (5)$$

$$\sum_{ki \in Y} \sum_{c \in C} Q_{kicd}^C - \sum_{ij \in Y} \sum_{c \in C} \left(Q_{ijcd}^C + I_{ijcd}^{\text{sqr}} x_c^C l_{ij} \right) + Q_{id}^S + Q_{igd}^G = \mu_d Q_i^L; \forall i \in B, d \in D \quad (6)$$

$$P_{id}^S = 0, \quad Q_{id}^S = 0; \forall i \notin B_S, d \in D \quad (7)$$

$$P_{igd}^G = 0, \quad Q_{igd}^G = 0; \forall i \notin B_G, d \in D \quad (8)$$

$$|P_{ijcd}^C| \leq \bar{V} \bar{I}_c^C y_{ijc}^C; \forall ij \in YY, c \in C, d \in D \quad (9)$$

$$|Q_{ijcd}^C| \leq \bar{V} \bar{I}_c^C y_{ijc}^C; \forall ij \in YY, c \in C, d \in D \quad (10)$$

$$I_{ijcd}^{\text{sqr}} \leq y_{ijc}^C (\bar{I}_c^C)^2; \forall ij \in YY, c \in C, d \in D \quad (11)$$

$$(P_{id}^S)^2 + (Q_{id}^S)^2 \leq \bar{S}_i^S; \forall i \in B_S, d \in D \quad (12)$$

$$(P_{igd}^G)^2 + (Q_{igd}^G)^2 \leq h_{ig}^G \bar{S}_g^G; \forall i \in B_G, g \in G, d \in D \quad (13)$$

$$-M(1 - y_{ijc}^C) \leq V_{id}^{\text{sqr}} - 2l_{ij} (r_c^C P_{ijcd}^C + x_c^C Q_{ijcd}^C) - (z_c^C)^2 l_{ij}^2 Z_{\text{base}} I_{ijcd}^{\text{sqr}} - V_{jd}^{\text{sqr}} \leq M(1 - y_{ijc}^C); \quad \forall ij \in Y, c \in C, d \in D \quad (14)$$

$$V_{jd}^{\text{sqr}} \tilde{I}_{ijcd}^{\text{sqr}} = \left(\tilde{P}_{ijcd}^C \right)^2 + \left(\tilde{Q}_{ijcd}^C \right)^2; \forall ij \in Y, d \in D \quad (15)$$

$$\tilde{I}_{ijcd}^{\text{sqr}} = \sum_{c \in C} I_{ijcd}^{\text{sqr}}; \forall ij \in Y, d \in D \quad (16)$$

$$\tilde{P}_{ijcd}^C = \sum_{c \in C} P_{ijcd}^C; \forall ij \in Y, d \in D \quad (17)$$

$$\tilde{Q}_{ijcd}^C = \sum_{c \in C} Q_{ijcd}^C; \forall ij \in Y, d \in D \quad (18)$$

$$V_{\min}^2 \leq V_{id}^{\text{sqtr}} \leq V_{\max}^2; \forall i \in B, d \in D \quad (19)$$

$$V_{id}^{\text{sqtr}} = 1; \forall i \in B_S, d \in D \quad (20)$$

$$\sum_{i \in B_g} \sum_{g \in G} P_{igd}^G \leq \rho_{\max} \mu_d \sum_{j \in B} P_j^L; \forall d \in D \quad (21)$$

C. Radiality Constraints

To guarantee radial topology, equations (22), (23) and (24) constrain number of upstream buses for each load bus, transfer bus and substation bus; respectively. If a cable is installed in a route, one bus of the route should be upstream of the other bus as constrained by (25). Constraints for fictitious flow are also added to avoid islands formation in the presence of DG and they can be found in [16], [17].

$$\sum_{j \in N_i} w_{ji}^R = 1; \forall i \in B_L \quad (22)$$

$$\sum_{j \in N_i} w_{ji}^R \leq 1; \forall i \notin B_L \quad (23)$$

$$\sum_{j \in N_i} w_{ji}^R = 0; \forall i \in B_S \quad (24)$$

$$w_{ji}^R + w_{ij}^R = \sum_{c \in C} y_{ijc}^C; \forall ij \in Y \quad (25)$$

For more details on the power flow and radiality constraints, the reader can refer to [8], [12], [13], [16], [18].

III. SHORT CIRCUIT CONSTRAINTS

A. Network Equations for Short Circuit At Each Bus

Cables short circuit capacity is designed based on symmetrical three-phase short circuit current. Constraints that govern circuit modeling for a fault at each bus are given in the following. For a symmetrical fault at each bus b , real and imaginary parts of the complex Kirchhoff's voltage law (KVL) equations for each candidate cable in each candidate route are given in (26) and (27); respectively. Complex KVL equations for candidate generators are represented by (28) and (29). In (28) and (29), each generator is represented by a voltage source and impedance consisting of its subtransient impedance and connecting transformer impedance referred to the medium voltage side. It should be noted that the use of big-M method with binary decision variables in (26)-(29) imposes these constraints only on the installed cables or DG units. Voltage drop due to substation transformer impedance is represented in (30) and (31). It is noted from the previous equations that, lines, generators and substation transformers are exactly modeled by their resistance and reactance without approximations. Constraints (32) and (33) represent real and imaginary equations for Kirchhoff's current law (KCL) during short circuit at each bus. Following (34) and (35), short circuit current can be supplied only from substations buses and DG candidate buses. In (36) - (39), with incorporation of binary decision variables, short circuit current can flow only

in routes with installed cables and installed DG units. It is also noted that (38) and (39) constrain short circuit current supplied by generators to the fourth quadrant of the complex plane. Decision variables are incorporated in (26)-(29) and in the limits in (36)-(39) because the circuit topology and DG allocation and size are not known in advance. Connection to ground at the faulted bus is forced by (40).

$$\begin{aligned} -M(1-y_{ijc}^C) &\leq V_{bi}^{\text{re}} - V_{bj}^{\text{re}} - l_{ij} \left(r_c^C I_{bij}^{\text{R, re}} - x_c^C I_{bij}^{\text{R, im}} \right) \\ &\leq M(1-y_{ijc}^C); \forall b \in B_{\text{SC}}, ij \in Y, c \in C \end{aligned} \quad (26)$$

$$\begin{aligned} -M(1-y_{ijc}^C) &\leq V_{bi}^{\text{im}} - V_{bj}^{\text{im}} - l_{sr} \left(r_c^C I_{bij}^{\text{R, im}} + x_c^C I_{bij}^{\text{R, re}} \right) \\ &\leq M(1-y_{ijc}^C); \forall b \in B_{\text{SC}}, ij \in Y, c \in C \end{aligned} \quad (27)$$

$$\begin{aligned} -M(1-h_{ig}^G) &\leq U - V_{bi}^{\text{re}} - r_g^G I_{bi}^{\text{G, re}} + x_g^G I_{bi}^{\text{G, im}} \\ &\leq M(1-h_{ig}^G); \forall b \in B_{\text{SC}}, i \in B_G, g \in G \end{aligned} \quad (28)$$

$$\begin{aligned} -M(1-h_{ig}^G) &\leq -V_{bi}^{\text{im}} - r_g^G I_{bi}^{\text{G, im}} - x_g^G I_{bi}^{\text{G, re}} \\ &\leq M(1-h_{ig}^G); \forall b \in B_{\text{SC}}, i \in B_G, g \in G \end{aligned} \quad (29)$$

$$U - V_{bi}^{\text{re}} = r_g^G I_{bi}^{\text{G, re}} - x_g^G I_{bi}^{\text{G, im}}; \forall b \in B_{\text{SC}}, i \in B_S \quad (30)$$

$$-V_{bi}^{\text{im}} = r_g^G I_{bi}^{\text{G, im}} + x_g^G I_{bi}^{\text{G, re}}; \forall b \in B_{\text{SC}}, i \in B_S \quad (31)$$

$$\begin{aligned} \sum_{k: ki \in Y} I_{bki}^{\text{R, re}} - \sum_{j: ij \in Y} I_{bij}^{\text{R, re}} + I_{bi}^{\text{S, re}} + I_{bi}^{\text{G, re}} &= 0; \\ \forall b \in B_{\text{SC}}, i \in B, i \neq b \end{aligned} \quad (32)$$

$$\begin{aligned} \sum_{s: ki \in Y} I_{bki}^{\text{R, im}} - \sum_{j: ij \in Y} I_{bij}^{\text{R, im}} + I_{bi}^{\text{S, im}} + I_{bi}^{\text{G, im}} &= 0; \\ \forall b \in B_{\text{SC}}, i \in B, i \neq b \end{aligned} \quad (33)$$

$$I_{bi}^{\text{S, re}}, I_{bi}^{\text{S, im}} = 0; \forall b \in B_{\text{SC}}, i \in B, i \notin B_S \quad (34)$$

$$I_{big}^{\text{G, re}} = 0, I_{big}^{\text{G, im}} = 0; \forall b \in B_{\text{SC}}, g \in G, i \in B, i \notin B_G \quad (35)$$

$$-M \sum_{c \in C} y_{ijc}^C \leq I_{bij}^{\text{R, re}} \leq M \sum_{c \in C} y_{ijc}^C; \forall b \in B_{\text{SC}}, ij \in Y \quad (36)$$

$$-M \sum_{c \in C} y_{ijc}^C \leq I_{bij}^{\text{R, im}} \leq M \sum_{c \in C} y_{ijc}^C; \forall b \in B_{\text{SC}}, ij \in Y \quad (37)$$

$$0 \leq I_{bi}^{\text{G, re}} \leq M \sum_{g \in G} h_{ig}^G; \forall b \in B_{\text{SC}}, i \in B_G \quad (38)$$

$$-M \sum_{g \in G} h_{ig}^G \leq I_{bi}^{\text{G, im}} \leq 0; \forall b \in B_{\text{SC}}, i \in B_G \quad (39)$$

$$V_{bb}^{\text{re}} = 0, V_{bb}^{\text{im}} = 0; \forall b \in B_{\text{SC}} \quad (40)$$

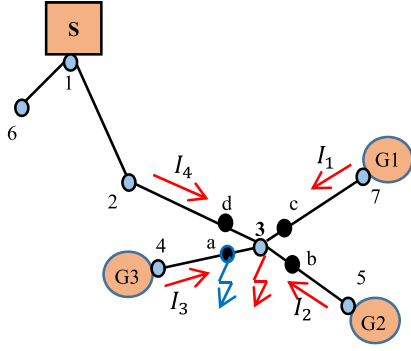


Fig. 1. Worst-case short circuit currents.

B. Worst-Case Short Circuit Current Calculation

For radial network topology, without DG, the worst-case short circuit current in a cable is calculated by simulating a short circuit at the upstream bus. For a short circuit in the cable near the upstream bus, the short circuit current flows through part of the cable and must be less than cable short circuit withstand capacity. For the case of DG, this should be done in the two ends of the cable due to contribution of DG in short circuit current. For example, in Fig. 1, if short circuit occurs at bus 3, substation (S) and the DGs contribute to fault current. The worst-case short circuit current in cables installed in routes connected to bus 3, i.e., (3,4), (3,5), (3,7) and (3,2) can be calculated by considering short circuit in the cables a small distance apart from bus 3 at points a , b , c and d ; respectively which gives:

$$\begin{aligned} I_{3,4}^{WC} &= I_1 + I_2 + I_4 & I_{3,5}^{WC} &= I_1 + I_3 + I_4 \\ I_{3,7}^{WC} &= I_2 + I_3 + I_4 & I_{3,2}^{WC} &= I_1 + I_2 + I_3 \end{aligned} \quad (41)$$

The worst-case current other ends of cables; $I_{4,3}^{WC}$, $I_{5,3}^{WC}$, $I_{7,3}^{WC}$ and $I_{2,3}^{WC}$ can be calculated by considering short circuit at buses 4, 5, 7 and 2; respectively. Thus, after considering short circuit at all network buses, the worst-case current in all candidate routes are calculated in both ends. Generalized equations for real and imaginary components of worst-case current associated with each short circuit bus are given in (42) and (43); respectively.

$$\begin{aligned} I_{bi}^{WC, re} &= \sum_{\substack{k: kb \in Y \\ k \neq i}} I_{bkb}^{R, re} - \sum_{\substack{j: bj \in Y \\ j \neq i}} I_{bbj}^{R, re} \\ &+ I_{bb}^{S, re} + I_{bb}^{G, re}; \forall b \in B_{SC}, i \in N_b \end{aligned} \quad (42)$$

$$\begin{aligned} I_{bi}^{WC, im} &= \sum_{\substack{k: kb \in Y \\ k \neq i}} I_{bkb}^{R, im} - \sum_{\substack{j: bj \in Y \\ j \neq i}} I_{bbj}^{R, im} \\ &+ I_{bb}^{S, im} + I_{bb}^{G, im}; \forall b \in B_{SC}, i \in N_b \end{aligned} \quad (43)$$

C. Cables Short Circuit Withstand Limits

Worst-case Current must be less than the installed cables short circuit withstand-capacity which is not known in advance.

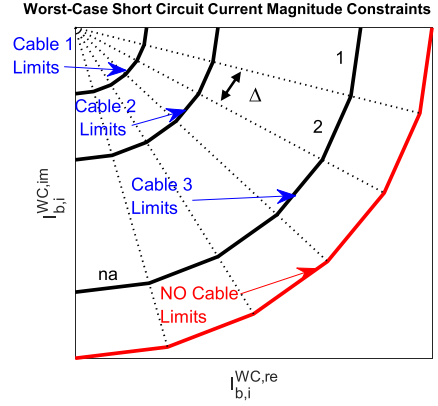


Fig. 2. Linearization of cables short circuit capacity magnitude.

Assuming zero phase angle for pre-fault voltage and considering characteristics of cables impedance, it can be shown that the vectors of worst-case currents lie in the fourth quadrant of the complex plane and for the cables to be adequately sized, these vectors must lie within the circle with radius equals the short circuit capacity of the corresponding cables sizes. In order to linearize these inequalities for MILP formulation, the circle in the fourth quadrant is linearized using n_v linear segments as shown in Fig. 2. Using start and end points of each segment, slopes of the segments are given by (44).

$$\begin{aligned} m_v &= \frac{\sin(v\Delta) - \sin((v-1)\Delta)}{\cos(v\Delta) - \cos((v-1)\Delta)}, \\ \Delta &= \frac{\pi/2}{n_v}; \forall v = 1, \dots, n_v \end{aligned} \quad (44)$$

Thus, worst-case magnitude inequality constraints associated with one end of each candidate route can be formulated as (45). In (45), the worst-case short circuit current is forced to the region determined by cable cross-sectional area in the fourth quadrant by equivalent segments inequality constraints. The equation of the other end (j) can be written in a similar way. Equation (45) is an extension of what has been proposed for branch currents in [8] but with the inclusion of decision variables which are unknown during solution. Magnitude of short circuit capacity depends on binary decision variables of the candidate cables.

$$\begin{aligned} I_{ij}^{WC, im} &\geq m_v I_{ij}^{WC, re} - (m_v \cos((v-1)\Delta) \\ &- \sin((v-1)\Delta)) \\ &\left(\sum_{c \in C} \bar{I}_c^{SC} y_{ijc}^C + \bar{I}_0^{SC} y_{ij0}^C \right); \forall ij \in Y, v = 1, \dots, n_v \end{aligned} \quad (45)$$

Thanks to the inclusion of “no cable” decision variable y_{ij0}^C , if no cable is selected for a certain route ij i.e. $y_{ij0}^C = 1$, the short circuit capacity of this route takes a value equals to the highest short circuit current in the system (short circuit is at substation bus) $\bar{I}_0^{SC}$ (according to (45)), so short circuit constraints are relaxed for this route segment.

Alternatives for candidate cables for each cable route, and for candidate generators are governed by (46) and (47); respectively.

$$\sum_{c \in C} y_{ijc}^C + y_{ij0}^C = 1; \forall ij \in Y \quad (46)$$

$$\sum_{g \in G} h_{ig}^G + h_{i0}^G = 1; \forall i \in B_G \quad (47)$$

D. Unbalanced Distribution System

The abovementioned short circuit equations assume that the used cables are multicore cables so the distribution system is balanced in the sense that mutual impedances between phases are equal and the system is represented by positive sequence impedances. These equations are valid also for single-core cables if they are arranged in trefoil formation. For single-core cables arranged in flat formation, mutual impedances are not equal. Consequently, the system can only be represented in three-phase coordinates. For example, equation (26) must hold for each phase and mutual resistance/reactance among phases must be included as given in (48). Equations (27)-(31) are modified in the same way. All other equations, (32) - (45), must hold for each phase.

$$\begin{aligned} -M(1 - y_{ijc}^C) &\leq [V_{bi}^{\text{re}}]_{\phi} - [V_{bj}^{\text{re}}]_{\phi} \\ -l_{ij} \sum_{\tilde{\phi} \in \{a, b, c\}} &\left([r_c^C]_{\phi\tilde{\phi}} [I_{bij}^{\text{R, re}}]_{\tilde{\phi}} - [x_c^C]_{\phi\tilde{\phi}} [I_{bij}^{\text{R, im}}]_{\tilde{\phi}} \right) \\ &\leq M(1 - y_{ijc}^C); \forall b \in B_{SC}, ij \in Y, c \in C, \phi \in \{a, b, c\} \end{aligned} \quad (48)$$

IV. SOLUTION USING BENDERS DECOMPOSITION

Due to the large number of short circuit constraints proposed in Section (III), solution of the problem in full formulation form would be very complex. A Benders decomposition- based approach is suggested to tackle this problem. The master problem is a MILP which minimizes the objective function (1) subject to constraints (5)-(25) and (46)–(47) without considering SCCs. The master problem results in decisions for cables routing, design and DG investment which are then passed to slave subproblems. Each sub-problem is a feasibility problem that represents short circuit at a system bus. Putting short circuit constraints (26) - (45) of short circuit at each bus b in a vector form, adding slack variables to inequality constraints and minimizing sum of the slack variables in the objective function, a slave sub-problem in iteration t can be formulated as follows.

$$\text{Minimize } F_{bt} = \sum_{i \in N_b} \sum_{v=1}^{n_v} s_{vbtiv} \quad (49)$$

$$A_{b1} X_{bt}^{\text{co}} - S V_{bt} \leq A_{b3} - A_{b2} \hat{X}_t^{\text{di}} : \lambda_{bt}^{\text{IN}} \quad (50)$$

$$A_{b4} X_{bt}^{\text{co}} = A_{b6} - A_{b5} \hat{X}_t^{\text{di}} : \lambda_{bt}^{\text{EQ}} \quad (51)$$

Where X_{bt}^{co} is the vector of continuous slave variables and $\hat{X}_t^{\text{di}}$ is the vector of discrete variables decided by the master problem at iteration t . If the objective function of any slave

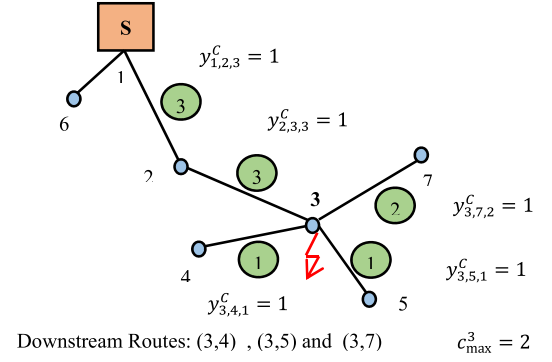


Fig. 3. Combinatorial cuts of infeasible subproblem of short circuit at bus 3.

sub-problem gives a non-zero value, Benders cut is added to the master problem at the next iteration as follows [11].

$$\left[(\lambda_{bt}^{\text{IN}})^T A_{b2} + (\lambda_{bt}^{\text{EQ}})^T A_{b5} \right] X^{\text{di}} \geq (\lambda_{bt}^{\text{IN}})^T A_{b3} + (\lambda_{bt}^{\text{EQ}})^T A_{b6} \quad (52)$$

V. PROPOSED COMBINATORIAL CUTS

The solution of master problem is degenerate [19] and Lagrange multipliers not unique. This may weaken Benders cuts and slow convergence. Combinatorial cuts, in conjunction with conventional Benders cuts can speed up convergence [20] by tightening solution space. Combinatorial cuts are generated depending on the logic combinations of the problem variables to exclude infeasible solutions from the master problem solution space. In the problem under study, for example, Fig. 3 shows subproblem of short circuit at bus 3 with no DG in a simple radial topology. Cables sizes are shown in circles. Suppose that worst-case currents of all cables downstream of bus 3, namely (3,4), (3,5) and (3,7) with sizes 1, 1, and 2; respectively violate their short circuit capacity. The maximum violated downstream cable size is size 2. Therefore, to exclude infeasibility in upcoming iterations, if all cables installed in routes upstream to bus 3 remain the same, all cables installed in routes downstream to bus 3 must be either greater than size 2 or not installed. Assuming that maximum available cable size is 3, this can be achieved by adding the following inequality for downstream route (3,4):

$$(1 - y_{1,2,3}^C) + (1 - y_{2,3,3}^C) + y_{3,4,3}^C + y_{3,4,0}^C \geq 1 \quad (53)$$

Similar equations are also added for the other two cables with violated constraints. For a slave subproblem with DG, if infeasible, decision variables of DG upstream of the short circuit bus are added to cut equations. Generalization of combinatorial cuts is given in (54) where $c_{b,\max}$ is the maximum cable size with violated constraints downstream of bus b .

$$\begin{aligned} \sum_{\alpha\beta \in UR_b} \sum_{\tilde{c} \in C, \tilde{y}_{\alpha\beta\tilde{c}}^C = 1} (1 - y_{\alpha\beta\tilde{c}}^C) + \sum_{k \in UB_b} \sum_{\tilde{g} \in G, \tilde{h}_{k\tilde{g}}^G = 1} (1 - h_{k\tilde{g}}^G) \\ + \sum_{c > c_{b,\max}} y_{ijc}^C + y_{ijo}^C \geq 1; \forall ij \in DR_b \end{aligned} \quad (54)$$

TABLE I
DATA FOR CANDIDATE CABLES AND DGs

	Cable 1	Cable 2	Cable 3		Gen 1	Gen2
	Al /XLPE					
Area (mm ²)	3x35	3x120	3x300	$\bar{S}_g^U$	1000	2000
r_c^C	1.1126	0.4100	0.1300	$\%x''$	18%	18%
x_c^C	0.1315	0.1121	0.0939	r/x''	0.15	0.15
$\bar{I}_c^C$	0.118	0.203	0.384	$\%z^T$	4%	6%
$\bar{I}_{c,SC}^C$	3.3	8.9	28.1	$\%r^T$	1%	1%
$C_g^{C,1}$	12500	17900	35500	$C_g^{G,1}$	500	490
$C_g^{C,M}$	450	450	450	$C_g^{G,M}$	22500	44100
				$C_g^{G,E}$	0.047	0.045

For problems involving DG allocation, to explore the minimum infeasible subspace, a first slave subproblem is solved with exclusion of DG. This is because the combination of cables may cause infeasibility with short circuit current supplied from substation only. If first slave subproblem is feasible, a second slave sub-problem is solved including DG.

VI. NUMERICAL RESULTS AND DISCUSSION

For numerical application, short circuit level at substation buses is 500 MVA at 11 kV with 0.1 R/X ratio. All buses are included in the set of short circuit buses. First, it is assumed that all candidate generators are 690 V synchronous-based generators connected to the grid through 690V/11 kV transformers with the same generators' ratings. Data for candidate generators subtransient reactance x'' , resistance r , connecting transformer resistance r^T and impedance z^T is listed in Table I. Other technical and economic data of distributed generation sizes are also given in Table I [16]. Rated power factor of generators is 0.9 lagging.

Number of segments for linearization of short circuit magnitude is 20. Project lifetime is 15 years and interest rate is 10%. Pre-fault voltage is 1.1 per unit as recommended by the IEC standard for 11 kV [21]. Two load levels are considered with load factors 0.7 and 1, annual durations 7760 and 1000 hours and cost of energy from substations 0.0577 and 0.0853 USD/kWh; respectively [16]. Minimum and maximum allowable voltage levels are 0.95 and 1.05 per unit; respectively. Maximum allowed DG penetration level is 20%. Master problem and Slave problems are solved using CPLEX 12.8 [22] with default settings. Three systems are used for numerical simulation; the 54-Bus System has 4 substations and 63 candidate routes, the 86-Bus System has 3 substations and 94 candidate routes and the 138-Bus System has 3 substations and 151 candidate routes. For all systems, maximum Power available at each substation is 5 MVA. Nodal power values and route lengths for all systems can be found in [18]. Total power is 18 MVA for 54 bus system and 13 MVA for the other systems all at 0.9 lagging power factor. Three candidate cable sizes are used, their technical and economic data is given in Table I. Cables of sizes 1, 2 and 3 are used in the 54 bus system while sizes 2 and 3 are used in the other systems. Candidate buses for distributed generation can be found in [18].

Table II gives the optimal costs of all cases with/without DG, with/without SCCs and upgraded cases. The upgraded network is obtained by solving the optimal distribution planning

TABLE II
NUMERICAL RESULTS OF TEST SYSTEMS (COST IN 10⁶ USD)

System	54 Bus		86 Bus		138 Bus	
DG	No	Yes	No	Yes	No	Yes
Without SCCs						
Invest., Loss. and Maint.	0.24	0.40	0.17	0.32	0.25	0.41
DG Energy	N/A	0.71	N/A	0.65	N/A	0.65
Substation Energy	6.46	5.49	4.67	3.78	4.67	3.78
Total	6.69	6.61	4.83	4.75	4.91	4.84
DG Bus(Size)	N/A	33(2)	N/A	78(2)	N/A	108(2)
With SCCs						
Invest., Loss. and Maint.	0.30	0.52	0.22	0.37	0.31	0.46
DG Energy	N/A	0.92	N/A	0.65	N/A	0.65
Substation Energy	6.46	5.21	4.67	3.78	4.67	3.78
Total	6.76	6.65	4.88	4.80	4.98	4.89
DG Bus(Size)	N/A	8 (2) 38(1)	N/A	33(2)	N/A	31(2)
Computation						
No. of SCCs (FS)	63468	66718	116452	147472	295616	312196
Cuts (BDS)	338	451	542	737	925	916
Execution Time (S)	23297	24417	9370	8303	12716	12002
Iterations	9	21	16	43	15	17
Upgraded Network						
Invest., Loss. and Maint.	0.34	0.56	0.24	0.39	0.36	0.49
% Reduction	11.76%	7.14%	8.33%	5.13%	13.89 %	6.12%

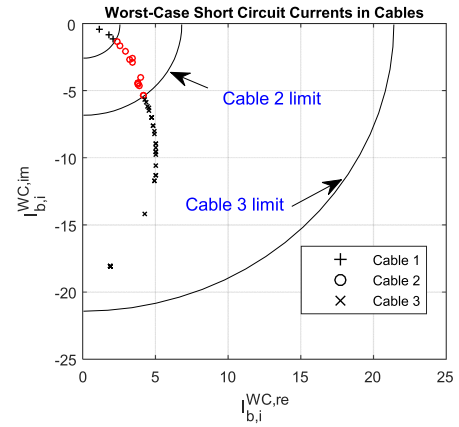


Fig. 4. Worst-case SC current vectors for 54-bus sys. with SCCs no DG.

problem without considering short circuit constraints and then performing short circuit study on the obtained network solution starting from upstream to downstream. If any cable section worst-case short circuit current exceeds the maximum short circuit withstand capacity decided by its cross-sectional area in the solution, the cross-sectional area is increased size by size until the short circuit withstand capacity is adequate. Then the evaluation process moves to the next downstream cable taking into consideration the new upstream sizes and so on. In cases without DG, it is sufficient to move one sweep from substation until covering all downstream buses. However, for the cases with distributed generation, it is necessary to repeat the whole process again until all cross-sectional areas are adequate. This is because, in the presence of DG, changing the cross-sectional area of any cable section may change the short circuit current for upstream sections as well.

Starting with the cases without DG, Fig. 4 shows vectors locations of the worst-case currents of the cables and their optimal sizes for 54 bus system. It is clear that cables are optimized with the minimum sizes that satisfy the short circuit capacity constraints at the worst-case condition. The optimal network

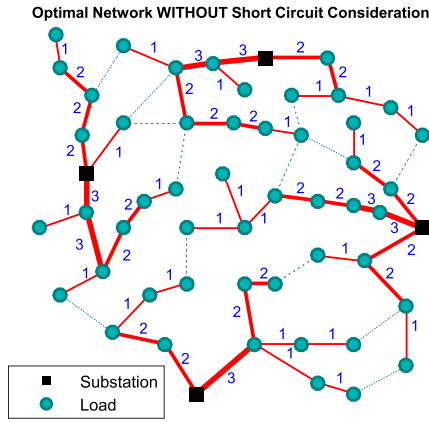


Fig. 5. Optimal solution for 54-bus system without SCCs, no DG.

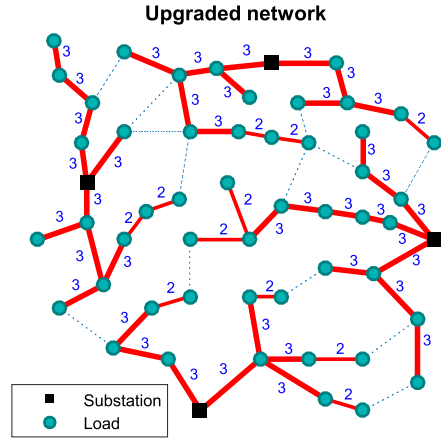


Fig. 7. Upgraded network for Fig. 5.

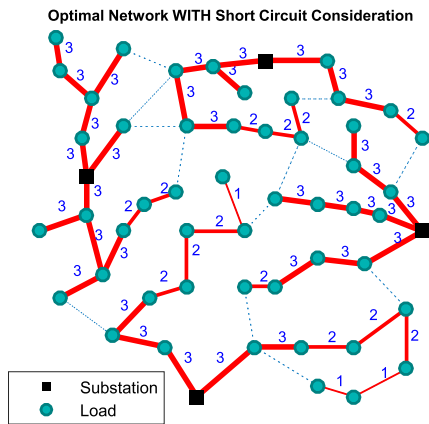


Fig. 6. Optimal solution for 54-bus system with SCCs no DG.

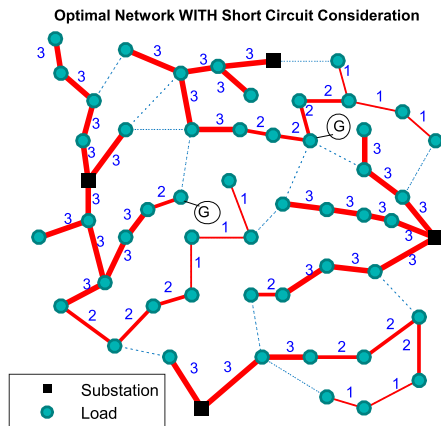


Fig. 8. Optimal solution for 54-bus system with SCCs with DG.

routing and cable sizes are shown in Fig. 5 for the case without SCCs and in Fig. 6 for the case that considers SCCs. The red lines indicate the routes with decision to install cables. The thickness of the red solid lines indicates the decided size of the installed cable and the size is also written beside each route (either size 1, 2 or 3). On the other hand the blue dotted lines indicate the routes with decision not to install any cables. It is clear that not only cable sizes are different but even cables routing is different in both cases. The network in Fig. 5 is upgraded to satisfy SCCs constraints with the same topology and the resulting cable sizes are given in Fig. 7.

Comparing the cost of investment, maintenance and losses for the two cases reveals that considering SCCs in the optimization problem has resulted in 11.76% saving in 54-bus system. Savings in 84-bus system and 138-bus system are 8.33% and 13.89%; respectively with no DG. For the cases with DG, savings are 7.14%, 5.13%, and 6.12%; respectively. Fig. 8 shows the optimal routing and cable sizes for the case with DG for the 54 bus system. The optimal locations of DG are at bus 8 with size 2 and at bus 38 with size 1. Per unit buses' voltages at the two load levels for this case are shown in Fig. 9. All bus voltages are within the allowable limits. In some buses, voltage at level 2 is greater than level 1 due to more DG power injection at downstream buses at level 2.

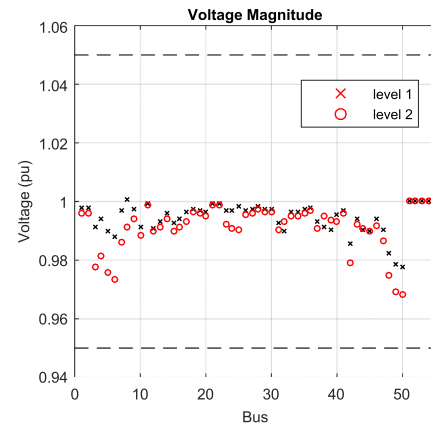


Fig. 9. Voltage profile for optimal solution of 54-bus system with SCCs with DG.

Fig. 10 shows the optimal network routing and cable sizes for the 138 bus system with DG considering SCCs. The optimal DG location is at bus 31 with size 2. Voltages of buses are all within limits in both load levels as shown in Fig. 11. For the 138 bus system, for the case without DG and considering SCCs constraints, all cables are sized to the minimum size that satisfies SCCs except cables in routes (93,94) and (106,94). Both

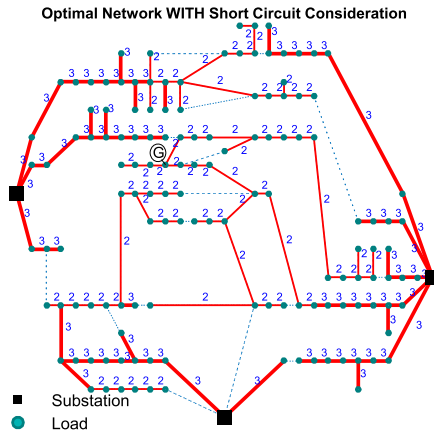


Fig. 10. Optimal solution for 138-bus system with SCCs with DG.

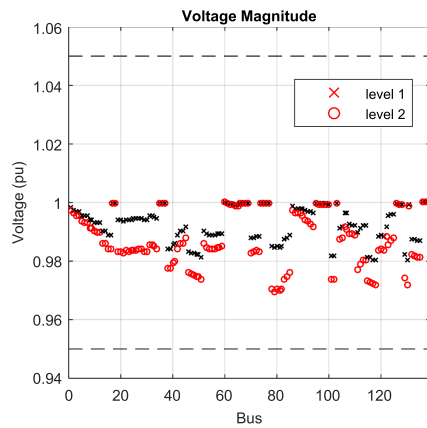


Fig. 11. Voltage profile of the optimal solution for 138-bus system with SCCs with DG.

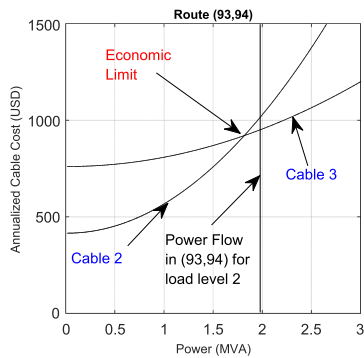


Fig. 12. Annualized cost of cables of route (93-94) of 138-bus system.

cables are of size 3 unless their worst-case currents lie in size 2 capability region. The reason can be explained by investigating Fig. 12 which shows the annualized investment, maintenance and losses cost of the two cable sizes at the route (93,94) as function of loading. It is found that load at this cable exceeds its maximum size 2 economical cost.

Table II shows comparison between number of SCCs added to the master problem between full solution (FS) and Benders decomposition solution (BDS). The execution time and iterations mentioned in Table II is for the Benders Decomposition

TABLE III
RENEWABLE GENERATION SCENARIOS

Generation Scenario	1	2	3	4
Maximum Power	0%	20%	50%	100%
Probability	0.75	0.07	0.08	0.10

TABLE IV
EFFECT OF RENEWABLE (INVERTER-BASED) GENERATION (COST IN 10^6 USD)

System	54-Bus		86-Bus		138 Bus	
Renewable DG Included	No	Yes	No	Yes	No	Yes
Without SCCs						
Investment losses and maintenance	0.403	0.463	0.315	0.384	0.415	0.473
Total	6.611	6.522	4.747	4.697	4.845	4.788
Conventional DG Bus (Size)	33(2)	8(2)	78(2)	54(2)	31(2)	120(2)
Renewable DG Bus (Size)		24(2) 43(1)		6(1) 34(2)		10(1) 38(2)
With SCCs						
Investment losses and maintenance	0.517	0.583	0.369	0.445	0.466	0.521
Total	6.653	6.603	4.801	4.76	4.897	4.851
Conventional DG Bus (Size)	8(2) 38(1)	31(2)	33(2)	44(2)	31(2)	114(2)
Renewable DG Bus (Size)		4(1) 12(1) 24(1) 43(2)		45(1) 70(2)		38(1) 64(2)
% Reduction of Investment, losses and maintenance cost With respect to the upgraded case	7.14	3.36	5.14	2.11	6.12	2.17

Solution (BDS). Solution of the master problem and all slave subproblems is considered one iteration. It is clear that BDS has effectively reduced number of constraints added to master problem.

The proposed optimization model can also accommodate renewable inverter-based distributed generation. Two photovoltaic (PV) generator alternatives of sizes 0.7 and 1.65 MVA are used. Technical and economic data of the PV generators as well as candidate buses are imported from [18]. As an inverter-based DG, the contribution of the renewable DG units in short circuit is limited to three times their full load current.

In order to account for the stochastic behavior of renewable-based DG, stochastic programming (SP) is used by considering multiple operating scenarios for the renewable DG. Losses and energy cost during all scenarios are weighted by their corresponding probabilities in the objective function. Also, all operational equality and inequality constraints are imposed for each scenario.

Four scenarios are considered for renewable generation with maximum power and probabilities as given in Table III. As two load levels are considered and weighted in the objective function by their duration, the total equivalent scenarios is eight. Results for the case when both synchronous-based and inverter-based generators are considered are presented in Table IV for all test systems. From the table it is found that the total cost for the cases with both synchronous-based and inverter-based DG units are less than the cases of synchronous-based DG units only with and without short circuit constraints. It is noticed also that saving with respect to upgrading method is less for

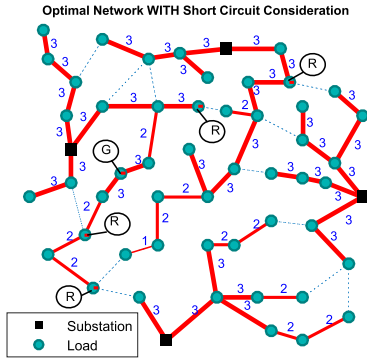


Fig. 13. Optimal solution for 54-bus system with SCCs with renewable and conventional DG.

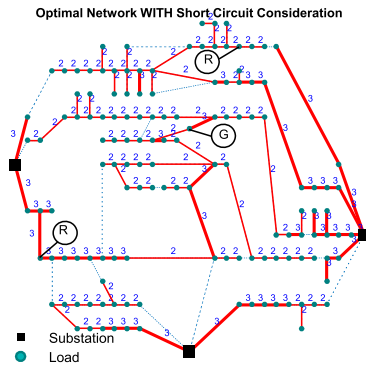


Fig. 14. Optimal solution for 138-bus system with SCCs with renewable and conventional DG.

TABLE V
EFFECT OF MAXIMUM DG PENETRATION LEVEL.(COST IN 10^6 USD)

System	54-Bus			86-Bus			138 Bus		
% DG Penetration Level	20	30	40	20	30	40	20	30	40
Without SCCs									
Investment losses and maintenance	0.403	0.541	0.602	0.315	0.397	0.48	0.415	0.489	0.564
Total	6.611	6.519	6.453	4.747	4.723	4.677	4.845	4.816	4.761
DG Bus (Size)	33(2)	8(2) 33(2)	8(2) 31(1) 33(2)	78(2)	54(2) 78(1)	10(2) 78 (2)	31(2)	31(2) 120(1)	31(2) 120(2)
With SCCs									
Investment Losses and maintenance	0.517	0.603	0.685	0.369	0.45	0.529	0.466	0.527	0.611
Total	6.653	6.581	6.536	4.801	4.777	4.725	4.897	4.854	4.808
DG Bus (Size)	8(2) 38(1)	8(2) 31(2)	8(2) 33(1) 38(2)	33(2)	44(1) 54(2)	33(2) 48(2)	31(2)	31(2) 78(1)	31(2) 120(2)
% Reduction 1	7.14	5.33	2.69	5.14	4.66	4.15	6.12	5.37	4.98
% Reduction 2	14.65	10.93	6.04	10.48	9.69	11.64	14.65	17.73	15.79

% Reduction 1: Reduction of Investment, losses and maintenance cost with respect to the upgraded case
 % Reduction 2: Reduction of Cables Investment, losses and maintenance cost with respect to the upgraded case

cases with renewable sources due to less severe condition from short circuit point of view. Optimal solutions for 54-bus and 138-bus systems with SCCs with both synchronous-based and inverter-based candidate DG units are shown in Figs 13 and 14; respectively. Renewable generators are marked by letter “R”.

To validate the proposed optimization for higher DG penetration levels, maximum penetration levels of 20%, 30% and 40% are considered with synchronous-based DG and the results are given in Table V. In all cases the proposed model with

TABLE VI
EFFECT OF DG UNITS OWNED BY OTHER PARTIES(COST IN 10^6 USD)

System	54-Bus		86-Bus		138 Bus	
DG Investment Included	Yes	No	Yes	No	Yes	No
Without SCCs						
Investment losses and maintenance	0.403	0.302	0.315	0.199	0.415	0.317
Total	6.611	6.407	4.747	4.631	4.845	4.729
DG Bus (Size)	33(2)	8(2) 31(2)	78(2)	78(2)	31(2)	31(2) 114(1)
With SCCs						
Investment losses and maintenance	0.517	0.37	0.369	0.29	0.466	0.378
Total	6.653	6.475	4.801	4.701	4.897	4.788
DG Bus (Size)	8(2) 38(1)	8(2) 33(2)	33(2)	42(1) 45(2)	31(2)	31(2) 78(1)
% Reduction of Investment, losses and maintenance cost with respect to the upgraded case	7.14	7.26	5.14	4.69	6.12	7.57

SCCs attained lower investment, losses and maintenance cost compared to the upgrading method.

In case of availability of existing distributed generation owned by other parties, investment cost of these distributed generators is not considered and only their energy cost is included in the objective function beside the losses cost and cables investment cost. Table VI presents the results of this situation for the three test systems for 20% maximum penetration level. Considering SCCs has resulted in lower investment, losses and maintenance cost compared to the upgraded network.

It is a mandatory technical requirement that the short circuit withstand capacity of each installed cable section in the distribution system is greater than the worst-case short circuit current that may flow in any part of the cable. The results presented in this paper have shown that including short circuit constraints in mathematical formulation of distribution system planning optimization problem is crucial for obtaining the most economical solution. It has been shown that the solution obtained by augmenting short circuit constraints with all other technical constraints in one optimization problem is superior to the non-optimal solution obtained by firstly ignoring short circuit constraints and then applying the common practice heuristic cross-sectional area upgrading method to the obtained solution. The heuristic upgrading method only increases cables cross-sectional areas starting from upstream to downstream to comply with short circuit levels while the non-optimal decided cable routes, distributed generation locations and sizes remain fixed.

The saving in annualized investment, losses and maintenance cost using the proposed optimization compared to the upgrading method has reached approximately 14% in some of the considered cases as shown in Table II. Saving depends on several factors including route lengths and loading levels. It is logical to expect that saving will be higher for distribution layouts with lighter loading such as in rural distribution networks.

Stochastically speaking, when optimization is performed under uncertainty as in some of the cases considered in this paper, the optimal solution is the solution that gives the minimum *expected* value of the cost. Also the calculated saving represents the *expected* value of saving considering uncertainty and scenarios probabilities in these cases. By applying stochastic programming, it is guaranteed that the designed system satisfies

the technical constraints including cables loading and voltage drop in all scenarios. The pre-fault voltage at substation is assumed 110% of the nominal voltage as recommended by the IEC standard for medium voltage distribution networks to represent the worst-case and most severe operating condition from short circuit point of view.

Optimal planning with short circuit consideration can also be solved using heuristic techniques such as genetic algorithm (GA) and particle swarm optimization (PSO), by performing short circuit study at each bus in each candidate solution and assign high weight to infeasible solutions in the fitness function but the solution quality cannot be accurately evaluated especially in highly combinatorial problems such as the one in this study.

Regardless of the solution strategy, to the best of the author's knowledge, cables short circuit withstand capacity and worst-case short circuit constraints have not been formulated or included in previous distribution system planning publications neither by mathematical programming nor by heuristic programming.

With regard to the added complexity of mathematical formulation, it is worth emphasizing that the proposed problem is an investment problem not an operational problem which means that the optimization is performed off-line and computation time is not critical taking into consideration the economic advantage of reaching the global minimum. The availability of computational resources such as parallel processing, cloud computing and computationally efficient solvers can provide substantial reduction in execution time for such complex and highly combinatorial problems.

VII. CONCLUSION

Complete formulation of short circuit constraints is introduced in this research for optimal distribution system routing, cable sizing, DG allocation and sizing. A Benders decomposition-based approach is suggested to handle the large number of constraints. Combinatorial cuts are proposed to speed up convergence and tighten solution space by examining routes, cable sizes and DG combinations in each iteration. The proposed optimization framework and solution algorithm is applied to 54 bus, 86 bus and 138 bus systems. Both synchronous-based and inverter-based distributed generators have been considered. Results show the effectiveness of the proposed optimization framework with regard to saving in investment, maintenance and losses cost compared to upgrading method. Also from computational point of view, the proposed solution algorithm has saved greatly the number of constraints added to the original problem.

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